

Stage 6 — Remedies, label audit, external validation and the paper

Where the Shortcut Sits: supervisor meeting 6 of 6

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In one sentence. What to do instead of masking depends on what carries the shortcut: annotation-free erasure (U-MtE) repairs overlays that carry it in their pixels, label-based reweighting is needed when it is carried by correlates, and no remedy improves on masking uniformly on unaltered data; the image audit that validates our artifact labels is prepared for a clinician but not yet done; the dermoscopy result replicates on new patients; and the paper, with every number traced to the regenerated results, is ready for the supervisor's review.

Notation

$Y \in \{0, 1\}$	diagnosis (1 = melanoma / malignant / solid-component tumour / erosion).
$A \in \{0, 1\}$	artifact present (hair, sonographer caliper, capsule debris, chest tube).
ROI	region of interest (lesion, nodule, tumour, lesion box, lungs); drawn in green ; artifacts in red .
r	overlap $ \text{artifact} \cap \text{ROI} / \text{artifact} $. Trap A: $r \geq 0.5$ (in-ROI); Trap B: $r < 0.1$ (out-of-ROI).
Environments	training and correlated test $P(A=1 Y=1) = 0.9$, $P(A=1 Y=0) = 0.1$; reversed test 0.1/0.9; clean test: artifact independent of Y (real traps: half of each class; sweeps: absent).
ERM, mask	linear head on the frozen features of the original image / of the ROI-masked image.
Crossover	$[\text{mask} - \text{ERM}]_{\text{out}} - [\text{mask} - \text{ERM}]_{\text{in}}$ of reversed-test AUROC (real traps: Trap B – Trap A; sweeps and transplant: location interaction on the same images).
CI	95% crossed seed×image bootstrap interval (Stage 1, Sec. ??); every interval in these reports uses it.
$ \Delta p $	same-head counterfactual sensitivity: change of the predicted probability when the artifact is inserted.
SMD	standardised mean difference between two groups.

1 Where we are

Stages 1–5 established the location law, its mechanism, its causal checks and its clinical consequences. This last meeting covers what the thesis recommends in practice (the decision guide and the remedy tests of the primary family), the evidence still missing (a clinician's audit of the automatic artifact labels), the external validation that was feasible, the limitations, and the state of the paper: its structure, its number audit, the CLAIM checklist and draft submission statements.

2 Question

- Q6a** Which remedy should replace or complement masking, and does any of them transfer to unaltered data?
- Q6b** How reliable are the automatic artifact labels that define “inside” and “outside” — and how will a clinician check them?
- Q6c** Does the main result hold on new patients from a different collection?
- Q6d** Is the paper ready to submit, and what must the author still decide?

Design decisions The remedies were demoted from a headline method to a *decision guide* after the second review: they are tested in the primary family, but the paper’s claim is the location law, not a new method. The image audit is *blind* (neutral identifiers, cells hidden) and stratified by label so that precision and recall can be estimated per location cell. External validation was run only if a pre-registered gate held (direct download, licence, patient identifiers, masks, cell counts); datasets that need an agreement were listed, not used.

3 Methods

Remedies ROI masking; U-MtE (mask-then-erase: after masking, erase from the frozen features the subspace along which *generic* synthetic overlays move the representation of masked images; no artifact annotation); U-MtE protected (label directions protected from erasure); group balancing and DFR (need image-level artifact labels); U-MtE with balancing. Primary tests: P2 (protected U-MtE > mask), P3 (erasure > augmentation with the same overlays), P4 (U-MtE with balancing > balancing), Trap A, Holm over the 16-test family.

Image audit (Part B) 240 images: 4 cohorts $\times$ 3 cells (artifact inside the ROI, outside it, artifact-free, by our automatic labels) $\times$ 20 (10 positive, 10 negative diagnosis), fixed seed, the 518-px images the models saw. Each image and its overlay (ROI outline, automatic artifact mask) have a neutral identifier; the reviewer fills artifact presence, location, mask correctness and ROI correctness in `audit/review_sheet.csv`. The key is kept outside the repository and fixed by its SHA-256 (4b5db1521f1fc214...); `audit/analyse_audit.py` computes precision, recall, location agreement, ROI correctness and, with a second reviewer, Cohen’s κ , and writes Table 6 of this report.

External validation (A4) ISIC 2020 (Rotemberg et al., 2021): lesion masks from the existing U-Net (trained on ISIC 2018 only); hair/ruler masks from a segmenter trained only on the ISIC 2019 masks and frozen before any ISIC 2020 image was scored (hair-free balanced accuracy 0.929, pixel IoU 0.601); the ISIC 2019 trap protocol unchanged with patient-disjoint folds.

4 Experiments

Table 1: Experiments of this stage. Commit times come from this machine and are not independent proof of order.

Experiment	Status	Registration	First commit (local time)	Support criterion	
U-MtE and variants (P2–P4)	supported registered	UNIVERSAL_I2E	ee95f10, 20:14	2026-09-26	Holm $p < 0.05$, estimate > 0
Erasure-rank ablation	supported registered	UMTE_ABLATION	30f19c8, 14:15	2026-09-27	descriptive
16-test primary family	not supported grouped after the individual results	STATISTICAL_PLAN	4e9d44c, 09:37	2026-09-27	Holm over 16
Natural-data remedies	supported registered	REVIEW2	29cd0da, 18:56	2026-09-27	descriptive (decision guide)
Image audit (prepared, not run)	supported registered	FINAL (Part B)	c45d482, 00:35	2026-09-28	precision, recall, location agreement, κ
External validation X1	supported registered	FINAL (A4)	c45d482, 00:35	2026-09-28	gate, then crossover CI > 0

5 Results

5.1 Remedies and the decision guide

Table 2 and Figure 1 compare the remedies in the in-ROI trap. Where the artifact carries the shortcut in its pixels (calipers), U-MtE, which needs no annotation, repairs most of what masking leaves (thyroid $+0.333$ [$+0.302, +0.364$] versus ERM, masking $+0.109$ [$+0.085, +0.133$]); with image-level artifact labels, U-MtE with balancing is best ($+0.498$ [$+0.471, +0.524$]). Where the shortcut is carried by correlates of the artifact (dermoscopic hair), no pixel-level method helps (U-MtE -0.009 [$-0.043, +0.024$]) and reweighting is required (balancing $+0.213$ [$+0.183, +0.243$], DFR $+0.196$ [$+0.157, +0.235$]). Where the artifact resembles the disease (capsule debris versus erosion fibrin), unprotected erasure removes disease evidence (-0.042 [$-0.076, -0.009$]), protection repairs this ($+0.078$ [$+0.047, +0.107$]), and label-based methods are best ($+0.258$ [$+0.221, +0.293$]). In the primary family, 7 of 12 remedy tests are supported after Holm correction (Table 3).

Table 2: Remedies in the in-ROI trap: change in reversed-test AUROC versus ERM (DINOv2, 5 seeds $\times$ 5 folds, crossed CIs). U-MtE needs no artifact annotation; balancing and DFR need image-level artifact labels.

Cohort (Trap A, in-ROI)	ROI masking	U-MtE	U-MtE protected	group balancing	DFR	U-MtE + balancing
Dermoscopy hair	-0.049 [$-0.083, -0.018$]	-0.009 [$-0.043, +0.024$]	-0.008 [$-0.046, +0.028$]	$+0.213$ [$+0.183, +0.243$]	$+0.196$ [$+0.157, +0.235$]	$+0.185$ [$+0.145, +0.224$]
Thyroid calipers	$+0.109$ [$+0.085, +0.133$]	$+0.333$ [$+0.302, +0.364$]	$+0.393$ [$+0.347, +0.434$]	$+0.388$ [$+0.360, +0.418$]	$+0.463$ [$+0.436, +0.492$]	$+0.498$ [$+0.471, +0.524$]
Ovarian calipers	$+0.132$ [$+0.072, +0.196$]	$+0.152$ [$+0.098, +0.203$]	$+0.175$ [$+0.116, +0.235$]	$+0.290$ [$+0.248, +0.332$]	$+0.310$ [$+0.263, +0.357$]	$+0.321$ [$+0.271, +0.372$]
Capsule debris	$+0.084$ [$+0.055, +0.112$]	-0.042 [$-0.076, -0.009$]	$+0.078$ [$+0.047, +0.107$]	$+0.179$ [$+0.150, +0.205$]	$+0.234$ [$+0.199, +0.273$]	$+0.258$ [$+0.221, +0.293$]

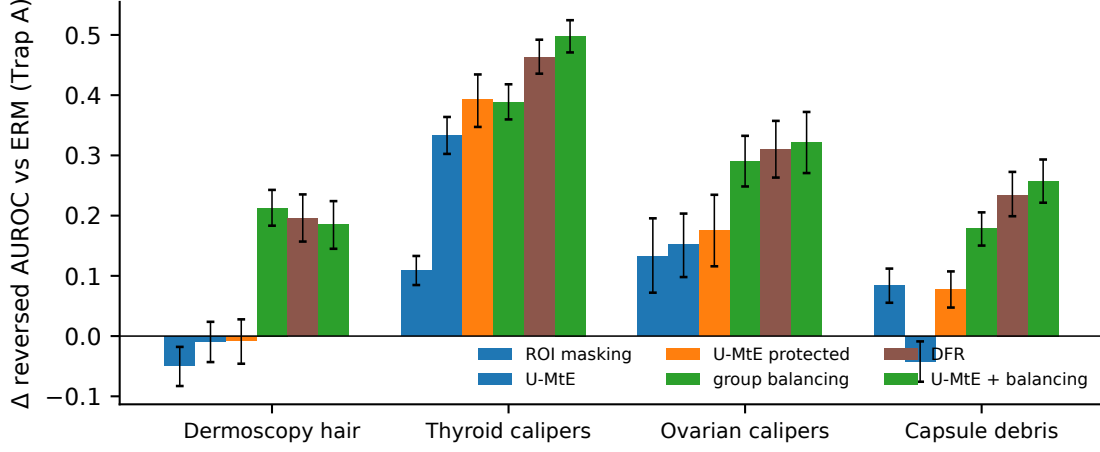


Figure 1: Remedies in the in-ROI trap (change in reversed AUROC versus ERM, crossed 95% CIs).

Table 3: Remedy tests of the 16-test primary family (Trap A, reversed test).

Family	Cohort	Estimate [95% CI]	p (Holm, 16)	Supported
P2 U-MtE (protected) > mask	ISIC hair	+0.042 [+0.024, +0.058]	0.0016	supported
P2 U-MtE (protected) > mask	Thyroid	+0.283 [+0.237, +0.329]	0.0016	supported
P2 U-MtE (protected) > mask	Capsule	-0.006 [-0.018, +0.006]	0.3486	not supported
P2 U-MtE (protected) > mask	Ovary	+0.043 [+0.023, +0.065]	0.0016	supported
P3 erase > augment	ISIC hair	+0.035 [+0.017, +0.055]	0.0016	supported
P3 erase > augment	Thyroid	+0.215 [+0.189, +0.244]	0.0016	supported
P3 erase > augment	Capsule	-0.139 [-0.158, -0.120]	0.0016	not supported (opposite)
P3 erase > augment	Ovary	+0.035 [+0.003, +0.067]	0.1304	not supported
P4 U-MtE+balanced > balanced	ISIC hair	-0.028 [-0.064, +0.007]	0.3486	not supported
P4 U-MtE+balanced > balanced	Thyroid	+0.110 [+0.079, +0.142]	0.0016	supported
P4 U-MtE+balanced > balanced	Capsule	+0.079 [+0.042, +0.115]	0.0016	supported
P4 U-MtE+balanced > balanced	Ovary	+0.031 [-0.014, +0.077]	0.3486	not supported

On unaltered data no remedy transfers uniformly On the natural test sets (Table 4), of 20 remedy–test-set comparisons with masking, 5 favour the remedy and 8 favour masking; the rest are not significant. The decision guide therefore tells the reader which remedy fits which regime and how to diagnose the regime (oracle artifact removal on a few annotated images; the theory’s fitted artifact–disease overlap), not which method to use everywhere.

Table 4: Remedies on the natural test sets (AUROC, all pairs, crossed CIs): no remedy improves on masking uniformly.

Natural test set	U-MtE – mask	U-MtE protected – mask	balancing – mask	U-MtE + balancing – mask
Thyroid (official split)	+0.004 [-0.012, +0.020]	-0.021 [-0.034, -0.008]	-0.017 [-0.064, +0.030]	-0.005 [-0.023, +0.013]
ISIC, BCN held out	+0.007 [+0.004, +0.010]	-0.019 [-0.024, -0.014]	+0.022 [+0.012, +0.031]	+0.010 [+0.006, +0.015]
ISIC, HAM held out	+0.013 [+0.009, +0.017]	-0.006 [-0.016, +0.004]	-0.032 [-0.049, -0.013]	+0.018 [+0.009, +0.028]
ISIC, MSK held out	-0.008 [-0.014, -0.002]	-0.003 [-0.012, +0.006]	-0.018 [-0.041, +0.004]	-0.016 [-0.024, -0.008]
Capsule (held-out frames)	-0.005 [-0.011, +0.001]	-0.023 [-0.029, -0.017]	-0.086 [-0.109, -0.065]	-0.008 [-0.013, -0.001]

Table 5: The decision guide (paper, Sec. “What carries the shortcut decides the fix”).

Regime	Example	Recommended	Diagnostic
Overlay carries the shortcut in its pixels	calipers, rulers, tubes	mask, then U-MtE (no labels) or U-MtE with balancing (labels)	oracle removal of the artifact recovers most of the gap
Artifact resembles the pathology	capsule debris vs. fibrin	mask plus DFR; protected erasure if no labels	large fitted artifact–disease overlap
Shortcut carried by correlates	dermoscopic hair, chest drains	DFR or group balancing (labels needed)	oracle removal recovers little
Encoder ignores the out-of-ROI artifact	chest tube with RAD-DINO	masking unnecessary	reversed $\approx$ clean AUROC at no overlap

5.2 The image audit

The package is in `audit/` (Figure 2 shows the kind of image the reviewer sees; the audit images themselves are not reproduced here, to keep the review blind). Licence decisions per dataset are in Table 7: images whose redistribution is not confirmed are kept outside the repository and regenerated from the original data by the sampling script. Reviewed rows so far: 0 of 240.

Table 6: Image audit of the automatic labels (to be filled).

Audit not yet performed. The table of precision, recall, location agreement, Cohen’s κ and ROI-mask correctness per cohort is written here by `audit/analyse_audit.py` once the review sheet has been filled.

Table 7: Licence decisions for publishing images (`audit/LICENCES.md`); unclear means not published.

Dataset	Licence (source)	Committed?	Attribution
ISIC 2018 Task 1/2, ISIC 2019 images (incl. HAM10000, BCN_20000, MSK)	CC BY-NC 4.0 (challenge.isic-archive.com/data)	Yes (raw images)	ISIC Challenge datasets 2018/2019; Tschandl et al. 2018 (HAM10000); Combalia et al. 2019 (BCN_20000); Codella et al. (MSK)
HAM10000 lesion segmentations	CC BY-NC 4.0 (Harvard Dataverse doi:10.7910/DVN/DBW86T)	Yes (as ROI outlines)	Tschandl et al. 2020
ISIC 2019 hair/ruler masks (Wegley et al., Scholars’ Mine doi:10.71674/man1-qa33)	no licence stated on the repository page	No — every overlay that draws these masks stays local	—
TN3K images, nodule masks; TNCD labels	not stated for the data; the GitHub repositories (TRFE-Net, ACL) license their *code* under MIT only	No — all thyroid images local	—
MMOTU OTU_2d	CC BY 4.0 (figshare doi:10.6084/m9.figshare.25058690)	Yes	Zhao et al. 2022, MMOTU
SEE-AI project dataset	CC BY 4.0 (Kaggle ‘capsule-olo/kyucapsule’, the project’s account)	Yes	Yokote et al. 2023/2024, SEE-AI project
Capsule expert contamination masks	CC BY 4.0 (figshare 27645021)	Yes	figshare 27645021
NIH ChestX-ray14	not used in the audit or example figures	—	—

5.3 External validation on new patients

The search is recorded in `stages/EXTERNAL_VALIDATION_SEARCH.md`. ISIC 2020 met every condition: direct download, CC BY-NC licence, patient identifiers, masks producible by our frozen models, and the cell-count gate (hair-free melanomas 53, Trap A melanomas 79, Trap B melanomas 231;

every cell at least 50). On 32,997 images of 2,056 patients the crossover is $+0.250$ [$+0.174, +0.327$]: masking harms with hair on the lesion (-0.064 [$-0.128, -0.002$]) and helps with hair beside it ($+0.186$ [$+0.118, +0.249$]) (Table 8). Thyroid cohorts with patient identifiers either lack nodule masks (ThyUS2Path) or require an agreement (TN-SCUI 2020; Stanford AIMI) and were not used.

Table 8: External validation on new patients (ISIC 2020; patient-disjoint folds; DINOv2; crossed CIs).

ISIC 2020, mask – ERM	reversed test	correlated test	clean test
Trap A (hair on the lesion)	-0.064 [$-0.128, -0.002$]	$+0.068$ [$+0.040, +0.099$]	$+0.027$ [$-0.011, +0.066$]
Trap B (hair beside it)	$+0.186$ [$+0.118, +0.249$]	-0.130 [$-0.160, -0.100$]	$+0.005$ [$-0.049, +0.056$]

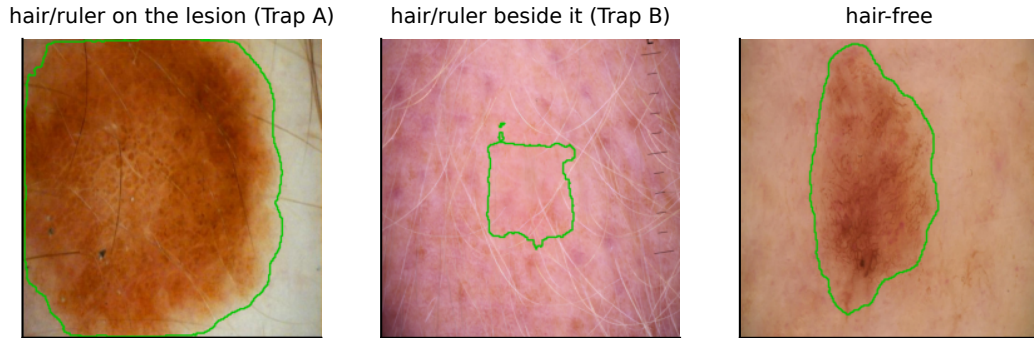


Figure 2: ISIC 2020 images from the external traps (CC BY-NC 4.0), with the lesion outline from the frozen U-Net (green); predicted hair/ruler masks are not drawn. Selection: a random image among the top quartile of predicted hair/ruler area in each artifact cell (seed 20260928); any hair-free image.

5.4 The paper

Title *Masking cannot remove what is inside the region of interest: location-dependent shortcut mitigation across dermoscopy, ultrasound and capsule endoscopy.* Target: *Medical Image Analysis*. Main text 22 pages (review format), supplement 18 pages; abstract 249 words (limit 250).

Structure Introduction; data and setup; the location law (controlled, real, transplant, matched, A2, replication including ISIC 2020, chest-radiograph boundary); mechanism; theory (held-out prediction); consequences on unaltered data (operating points, the post hoc thyroid subgroup labelled as exploratory confirmation); the decision guide; robustness; discussion (including “why this is not obvious” and eight limitations).

Number audit Every interval in the main text and supplement (the `\input` tree of `main.tex` and `supplement.tex`) must match an interval of a regenerated, crossed-bootstrap result file together with its estimate; every other three-decimal number must occur in a result file. Result: 367 intervals and 970 numbers checked, 0 failed to trace.

CLAIM 2024 43 items: 35 yes, 3 partial, 2 no, 3 not applicable (partial or no: items 13, 18, 21, 33, 36; checklist in the supplement). The remaining gaps are acquisition protocols the dataset providers did not release, the absence of external ultrasound and capsule cohorts, and the clinician audit.

5.5 Draft submission statements (for the author to confirm)

Data availability All datasets are public: ISIC 2018 and 2019 challenge data and ISIC 2020 (CC BY-NC 4.0), HAM10000 lesion masks (CC BY-NC 4.0), the published ISIC 2019 hair/ruler masks, TN3K with TNCD labels, MMOTU (CC BY 4.0), SEE-AI and its expert contamination masks (CC BY 4.0), NIH ChestX-ray14 and RANZCR-CLiP (competition terms). Derived masks, cohort files and all result files are in the repository; images are not redistributed where the licence does not allow it.

Code availability Code, pre-registrations with their commit history, result files and the scripts that regenerate every table, figure and number are in the public GitHub repository ([hossam1999/Where_shortcut_sits](#)).

Ethics Retrospective analysis of public, de-identified datasets released by their providers for re-search; no new data were collected and no patient was contacted. The author should confirm with the institution whether a formal waiver is required.

Use of AI assistance The experiments, analysis code, reports and parts of the manuscript text were produced with the assistance of an AI system (Claude, Anthropic) working under the author’s direction; the author designed the study, reviewed every result and is responsible for the content. The wording should follow the journal’s policy on generative-AI disclosure.

6 What failed or changed

- **The clinician audit has not been done.** The package is ready; until it is, artifact-label validity rests on the analysis agent’s visual audit (Stage 3).
- **No remedy transfers uniformly** to unaltered data; U-MtE was demoted from headline method to one option of the decision guide. Three remedy tests of the primary family changed with the corrected bootstrap (Stage 4).
- **External validation is dermoscopy only;** no thyroid cohort met the gate without an agreement or missing masks.
- **Chest-radiograph device and drain runs, LaMa inpainting and template erasure** could not be regenerated; they appear as point estimates labelled “not re-estimated”.
- **Figures committed before the licence check** ([report/figures/data_thyroid.png](#), [report/figures/data_isic.png](#)) conflict with the licence rule and need the author’s decision.

7 Conclusion

Supported. A decision guide grounded in what carries the shortcut; external replication of the dermoscopy crossover on new patients; a paper whose every interval traces to the corrected results. **Qualified.** The remedies help in the trap environments but none transfers uniformly to unaltered data. **Not yet available.** A clinician’s audit of the artifact labels; external validation in ultrasound and capsule endoscopy.

8 Questions for the supervisor

1. Who should perform the image audit (two dermatologists and a sonographer?), and should it be completed before submission?

2. Should the remedies stay in the paper as a decision guide, or move entirely to a second paper?
3. Is *Medical Image Analysis* the right target, given the methodological emphasis, or would a clinical-AI venue suit the operating-point results better?
4. Are the draft data-availability, ethics and AI-assistance statements acceptable to the institution?

9 Next stage

Before submission: (i) run the clinician audit and replace Table 6 with the audit analysis script (`audit/analyse_audit.py`); (ii) decide on the figures that conflict with the licence rule; (iii) confirm the submission statements; (iv) obtain the supervisor’s decisions above; (v) push the final registration record and tag the submitted version.

References

Veronica Rotemberg, Nicholas Kurtansky, Brigid Betz-Stablein, et al. A patient-centric dataset of images and metadata for identifying melanomas using clinical context. *Scientific Data*, 8:34, 2021.